

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 1: Information Representation

Subtopic 1.2.1: Multimedia: Graphics

Concept 1.2.1.1: Bitmap Image Encoding (Pixel, Header, Resolution, Colour Depth)

Total Questions: 16

Mark Schemes begin on Page 9

Question 1 | Source: 9618_w25_qp_11 (Q7.d.ii)

7 Web browsers use Internet Protocol (IP) addresses.

(d) A bitmap file is downloaded. The image has a maximum of 256 colours and measures 512 pixels wide by 2048 pixels high.

(ii) Explain why the actual file size may be larger than the one calculated in (d)(i).

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..... [2]

Question 2 | Source: 9618_s25_qp_11 (Q3.a)

3 A computer stores images and text files.

(a) One of the images is a bitmapped image.

Complete the table by writing the answer for each statement.

Statement	Answer
the term for the smallest element that makes up an image	
the largest number of different colours that can be represented with a bit depth of 8 bits	
the term for the dots per inch (dpi) when an image is displayed	

[3]

Question 3 | Source: 9618_s25_qp_13 (Q1.a.ii)

1 (a) A student paints a picture in an art class.

The student takes a photograph of the picture using a digital camera. The digital camera creates an image with a resolution of 2 million pixels and uses a bit depth of 16 bits.

- (ii) The student takes a second photograph with a lower bit depth.

Explain the effect of decreasing the bit depth on the image and on the image file.

Effect on image

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Effect on image file

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[4]

Question 4 | Source: 9618_w24_qp_12 (Q7.a.i)

- 7 A student takes a photograph of a science experiment.

- (a) The photograph is saved as a bitmapped image.

- (i) Define the following bitmap terms.

Colour depth

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File header

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[2]

Question 5 | Source: 9618_w24_qp_12 (Q7.a.ii)

- 7 A student takes a photograph of a science experiment.

- (a) The photograph is saved as a bitmapped image.

- (ii) Explain why changing the image resolution will affect the image quality and file size.

Image quality

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File size

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[2]

Question 6 | Source: 9618_w24_qp_13 (Q6.b.i)

- 6 A computer system stores text, images and sound.

- (b) The colour of each pixel in a bitmapped image is represented by 8 bits.

- (i) State the largest number of different colours that can be represented by 8 bits.

..... [1]

Question 7 | Source: 9618_w24_qp_13 (Q6.b.ii)

- 6 A computer system stores text, images and sound.

- (b) The colour of each pixel in a bitmapped image is represented by 8 bits.

- (ii) State **one** drawback of increasing the number of bits that represents each pixel in the bitmap image.

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..... [1]

Question 8 | Source: 9618_s24_qp_12 (Q2.d.i)

- 2 A computer game is being designed that users will be able to play using a virtual reality (VR) headset.

- (d) The computer can transmit a video made from bitmap images and vector graphic animations to the VR headset.

- (i) Describe how the data for a bitmapped image is encoded.

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..... [3]

Question 9 | Source: 9618_s24_qp_13 (Q2.c)

- 2 A photograph is stored as a bitmap image.

- (c) Identify **two** elements of a bitmap image that can be changed to reduce its file size.

1

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2

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[2]

Question 10 | Source: 9618_w23_qp_11 (Q1.b)

- (b) State what is meant by the **bit depth** of a bitmap image **and** explain how changing the bit depth affects the image.

Definition

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Explanation

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..... [3]

Question 11 | Source: 9618_w23_qp_13 (Q2.a)

- 2 (a) Describe the impact of increasing the image resolution on the quality of a bitmap graphic.

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..... [2]

Question 12 | Source: 9618_s23_qp_11 (Q1.a)

- 1 Images are being created to advertise holidays.
- Some of the images are bitmap images and some are vector graphics.

(a) Complete the table by defining the image terms.

Term	Definition
Drawing list	
Pixel	
Colour depth	

[3]

Question 13 | Source: 9618_s23_qp_11 (Q1.b.i)

- 1 Images are being created to advertise holidays.
- Some of the images are bitmap images and some are vector graphics.

(b) The bitmap images are photographs of the holiday locations.

(i) Colour depth and image resolution are both included in the file header of a bitmap image.

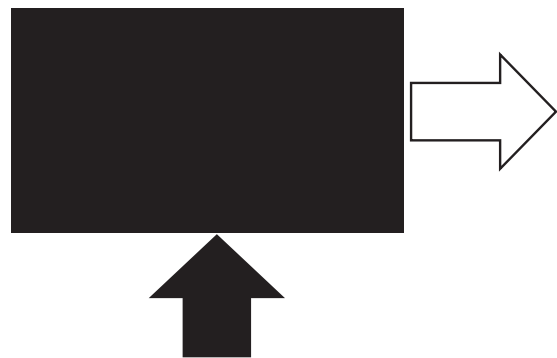
Identify **two other** items that could be included in the file header of each photograph.

- 1
- 2

[2]

Question 14 | Source: 9618_w22_qp_12 (Q8.a)

- 8 The following bitmap image has a resolution of 4096 × 4096 pixels and a colour depth of 24 bits per pixel.



The image is displayed on a monitor that has a screen resolution of 1920 × 1080 pixels.

(a) Tick (✓) **one** box in each row to identify the effect of each action on the image file size.

Action	Increases the file size	Decreases the file size	No change to the file size
Change the colour depth of the image file to 16 bits per pixel.			
Change the screen resolution to 1366 × 768 pixels.			
Change the colour of the rectangle from black to red.			

[2]

Question 15 | Source: 9618_w22_qp_13 (Q2.e)

- 2 The relational database `ASTRONOMY` is used to store data about telescopes, the companies that own the telescopes and the photographs taken by the telescopes.

The database has these three tables:

`COMPANY (TelephoneNumber, CompanyID, CompanyName)`

`PHOTOGRAPH (PhotoID, TelescopeID, DateTaken, TimeTaken, Elevation)`

`TELESCOPE (TelescopeID, CompanyID, SerialNumber)`

- (e) The photographs are stored as bitmap images.

Complete the statements about bitmap images by writing the missing words.

The of a bitmap image is the number of bits that are used to store each pixel.

Metadata about the image is stored in the of the file.

[2]

Question 16 | Source: 9618_s21_qp_11 (Q1.a.i)

- 1 Anya scans an image into her computer for a school project.

- (a) The scanned image is a bitmapped image.

- (i) Complete the following table to describe the two terms about graphics.

Term	Description
Pixel	
File header	

[2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q7.d.ii)

7(d)(ii)	1 mark per bullet point, max 2 marks - The file will contain metadata - The file will have a header	2
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Mark Scheme for Question 2 | Source: 9618_s25_ms_11 (Q3.a)

3(a)

1 mark for each correct answer

Statement	Answer
The term for the smallest element that makes up an image.	pixel
The largest number of different colours that can be represented with a bit depth of 8 bits.	256 // 2 ⁸
The term for the dots per inch (dpi) when an image is displayed.	Screen resolution

3

Mark Scheme for Question 3 | Source: 9618_s25_ms_13 (Q1.a.ii)

1(a)(ii)	1 mark each Image: - There will be fewer shades of colour available - so the image does not match the original as detail is lost Image file: - Fewer bits are used to store each pixel - so less data is stored, therefore the file size is reduced	4
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Mark Scheme for Question 4 | Source: 9618_w24_ms_12 (Q7.a.i)

7(a)(i)	1 mark for each correct definition Colour depth: - the number of bits used to represent a colour // the number of colours that can be represented in an image File header: - stores data about the image file / metadata	2
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Mark Scheme for Question 5 | Source: 9618_w24_ms_12 (Q7.a.ii)

7(a)(ii)	1 mark for each correct explanation Image quality: - Decreasing resolution means details within the image are lost because there are fewer pixels // Increasing resolution means the image is more detailed because there are more pixels File size: - Decreasing the resolution will decrease the file size because there are fewer pixels therefore less data // Increasing the resolution will increase the file size because there are more pixels therefore more data	2
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Mark Scheme for Question 6 | Source: 9618_w24_ms_13 (Q6.b.i)

6(b)(i)	1 mark for: 256 // 2^8	1
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Mark Scheme for Question 7 | Source: 9618_w24_ms_13 (Q6.b.ii)

6(b)(ii)	1 mark for: Increased file size	1
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Mark Scheme for Question 8 | Source: 9618_s24_ms_12 (Q2.d.i)

2(d)(i)	1 mark each: - Image is made of pixels and each pixel has one colour - Each colour has a unique binary code Code for the colour of each pixel is stored in sequence	3
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Mark Scheme for Question 9 | Source: 9618_s24_ms_13 (Q2.c)

2(c)	1 mark each to max 2 : - Colour/bit depth - Image resolution	2
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Mark Scheme for Question 10 | Source: 9618_w23_ms_11 (Q1.b)

1(b)	1 mark for the definition - The number of bits used to represent each colour 1 mark for each bullet point for the explanation - Increase in bit depth means the image has a greater range of colours // Decrease in bit depth means the image has a smaller range of colours - Increase in bit depth makes the image closer to the original / more realistic // Decrease in bit depth makes the image less like the original / less realistic	3
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Mark Scheme for Question 11 | Source: 9618_w23_ms_13 (Q2.a)

2(a)	1 mark for each bullet point: - more pixels can be stored / are available - the image is sharper / less pixelated	2
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Mark Scheme for Question 12 | Source: 9618_s23_ms_11 (Q1.a)

1(a)	1 mark for each correct definition	3								
<table><tr><th>Term</th><th>Definition</th></tr><tr><td>Drawing list</td><td>All the drawing objects in an image // a list that stores the commands required to draw each object</td></tr><tr><td>Pixel</td><td>The smallest part of the image // one square / dot of one colour</td></tr><tr><td>Colour depth</td><td>The number of bits per pixel // determines the number of colours that can be represented in the image</td></tr></table>			Term	Definition	Drawing list	All the drawing objects in an image // a list that stores the commands required to draw each object	Pixel	The smallest part of the image // one square / dot of one colour	Colour depth	The number of bits per pixel // determines the number of colours that can be represented in the image
Term	Definition									
Drawing list	All the drawing objects in an image // a list that stores the commands required to draw each object									
Pixel	The smallest part of the image // one square / dot of one colour									
Colour depth	The number of bits per pixel // determines the number of colours that can be represented in the image									

Mark Scheme for Question 13 | Source: 9618_s23_ms_11 (Q1.b.i)

1(b)(i)	1 mark each Examples: - Confirmation that it is a bitmap // file type - Compression type - Location/offset of data within the file - Dimensions e.g. 100 × 100 pixels	2
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Mark Scheme for Question 14 | Source: 9618_w22_ms_12 (Q8.a)

8(a)

1 mark one or two correct row(s). 2 marks for all three correct rows.

2

Action	Increases the file size	Decreases the file size	No change to the file size
Change the colour depth of the image file to 16 bits per pixel.		✓	
Change the screen resolution to 1366 × 768 pixels.			✓
Change the colour of the rectangle from black to red.			✓

Mark Scheme for Question 15 | Source: 9618_w22_ms_13 (Q2.e)

2(e)	1 mark for each correctly completed term; The <u>bit depth</u> of a bitmap image is the number of bits that are used to store each pixel. Metadata about the image is stored in the <u>header</u> of the file.	2
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Mark Scheme for Question 16 | Source: 9618_s21_ms_11 (Q1.a.i)

1(a)(i)	1 mark for each description Pixel: • A single square of one colour • The smallest addressable element in an image File header: • Data about the bitmap image (e.g. number of colours)	2
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